

测试分析报告

1 引言

1.1 编写目的

本报告记录 ROS2 智能药品拣选系统的自动化测试结果，评价 AI 医疗助手、认证授权、智能筛查、药品与库存、医生审批、订单流程及 ROS2 集成模块的当前质量，并为缺陷修复和后续 ROS2/Unity 联调提供依据。

预期读者为项目开发人员、测试人员、ROS2 集成人员及项目负责人。

1.2 被测对象

- 系统名称：ROS2 智能药品拣选系统
- 被测主体：`agent_with_backend`
- 测试范围：`tests/` 下默认可发现的自动化测试，以及显式执行的 `test-full-integration.py`
- 未测试范围：Unity、机器人硬件、安全渗透

1.3 测试环境

项目	实际环境
操作系统	Ubuntu 24.04
Python	3.10.11
pytest	9.0.3
Flask	3.1.3
SQLite	3.40.1
ROS2	相关测试使用 Mock/替身

测试通过 Flask 测试客户端和临时 SQLite 数据库执行。

1.4 术语

术语	定义
API	Application Programming Interface, 应用程序接口

术语	定义
JWT	JSON Web Token，接口认证令牌
ROS2	Robot Operating System 2，机器人操作系统第二版
E2E	End-to-End，端到端测试
Mock	用可控替身模拟外部依赖
xfail	pytest 中预期失败且持续跟踪的已知缺陷

1.5 参考资料

- [pytest、Flask、Python 官方文档](#)
- [tests/test_documents/测试执行手册.md](#)
- [tests/test_documents/测试计划.md](#)

2 测试概要

2.1 测试文件

```
tests
├── test_documents
│   ├── 测试计划.md
│   ├── 测试分析报告.md
│   └── 测试执行手册.md
├── unit          # 单元测试
├── integration   # 集成测试
├── ros_integration # ros 相关集成测试
└── agent         # agent 相关测试
```

2.2 测试结果

默认执行全量 pytest 测试共收集 442 项：

结果	数量	占比
通过	440	99.55%
失败	0	0.00%
预期失败 (xfail)	2	0.45%
合计	442	100.00%

不计 xfail 的实际执行通过率为 $440 / 440 = 100\%$ 。最近一次全量覆盖率测试耗时 91.24 秒。

2.3 分组结果

测试集	结果	耗时	结论
tests/unit	114 passed, 3 warnings	15.41 秒	通过
tests/integration	14 passed, 2 xfailed, 30 warnings	21.68 秒	无非预期失败, 存在 2 个已知缺陷
tests/ros_integration	141 passed	15.82 秒	通过
tests/agent	143 passed	8.84 秒	通过
根目录 API 测试	28 passed, 40 warnings	24.06 秒	通过

2.4 覆盖率

使用 `.coveragerc` 排除测试代码及配置中指定的非统计范围后，生产代码共有 8746 条语句，未覆盖 1302 条，语句覆盖率为 **85%**。

覆盖较高的代表模块包括：

模块	覆盖率
ros_integration/state_store.py	100%
common/utils/symptom_db_service.py	100%
api/upload_controller.py	100%
database/connection.py	100%
common/utils/text_utils.py	100%
agent/session/manager.py	99%
agent/memory/manager.py	98%
agent/planner/models.py	98%
agent/subagents/form.py	97%

模块	覆盖率
common/utils/validation.py	97%
ros_integration/error_handler.py	95%
ros_integration/message_adapter.py	95%
common/utils/http_client.py	94%
agent/engine/medical_agent.py	94%
agent/subagents/extractor.py	92%
api/order_controller.py	85%

覆盖不足的关键模块仍包括 `auth/blueprint.py` (57%) 、
`api/approval_controller.py` (55%) 、 `ros_integration/bridge.py` (61%) 、
`ros_integration/task_publisher.py` (61%) 和 `api/drug_controller.py` (69%) 。

3 测试结果及发现

3.1 测试1（UT-001：药品 API 基础操作单元测试）

本项测试对药品控制器的创建、删除、库存调整、药品查询、低库存查询和临期药品查询进行验证。

比较项目	动态输出要求	实际动态输出	比较结果
用例执行	所有单元测试均成功执行	单元测试集 114 passed, 3 warnings in 15.41s , 其中药品控制器 6 项基础用例全部通过	功能用例全部通过
创建药品	返回成功结果, 并在数据库中生成药品数据	创建用例通过, 临时 SQLite 数据库中生成对应药品记录	符合要求
查询药品	返回与输入条件匹配的药品数据	普通查询、低库存和临期查询用例全部通过	符合要求
库存调整	输出调整后的库存数据, 数据库同步更新	库存调整用例通过	符合要求

比较项目	动态输出要求	实际动态输出	比较结果
执行时间	单元测试执行时间处于可接受范围	114 项单元测试耗时 15.41 秒	符合要求

发现:

- 1. 药品控制器的六项基础操作均得到预期动态输出，未发现功能失败。
- 2. 单元测试产生 3 条 SQLAlchemy 2.0 弃用警告，来自旧式 `declarative_base()` 导入方式；该问题未影响功能结果，但应在后续升级中修正。

3.2 测试2 (IT-001 ~ IT-005: API 集成测试)

本项测试覆盖认证生命周期、角色权限、智能筛查、筛查历史、药品与库存 API，以及不同模块之间的数据传递。

比较项目	动态输出要求	实际动态输出	比较结果
用例执行	API 调用链路成功率不低于 95%	14 passed, 2 xfailed, 30 warnings in 21.68s	无非预期失败，但有 2 项已知缺陷
认证生命周期	注册、登录、验证、刷新、注销输出正确，注销后刷新失败	对应请求依次返回成功；令牌注销后刷新请求返回 401	符合要求
权限控制	患者、医生和管理员只能访问其权限范围内的接口	患者访问库存接口返回 403，患者审批操作返回 403，管理员可查询审计日志	符合要求
症状标准化	“头疼”“发烧”转换为“头痛”“发热”，未知症状单独返回	输出包含标准症状 头痛、发热，并将“未知症状”列入未匹配项	符合要求
智能筛查	返回不超过指定数量的候选药品，并按置信度降序输出	返回 1~3 条结果，首项置信度不低于末项，历史记录成功保存	符合要求

比较项目	动态输出要求	实际动态输出	比较结果
审计完整性	认证和业务操作生成可查询的审计数据	登录操作出现在审计日志中；审批自动下单缺少库存流水	部分符合

发现:

- 1. 认证、权限、筛查和历史记录之间的数据传递正确，主要 API 集成链路可用。
- 2. `test_approval_dispatch_writes_inventory_transaction` 为严格 xfail。实际输出表明审批自动下单会扣减库存，但没有生成 `inventory_transactions` 记录，库存审计链不完整，记为 DEF-INT-001。
- 3. `test_recent_ros_state_reports_connected` 为严格 xfail。生产代码仍使用 Python 3.10 无法直接解析的尾缀 `z` 时间戳，最新 ROS 状态经 HTTP 查询时可能输出 `connected=false`。

3.3 测试3（E2E-001：完整用药流程测试）

本项测试验证患者提交审批、医生处理、库存扣减、订单生成、任务发布调用和患者确认完成之间的完整业务流程。

比较项目	动态输出要求	实际动态输出	比较结果
患者提交审批	返回审批编号并保存待审批数据	请求返回 <code>201</code> ，生成 <code>approval_id</code>	符合要求
医生待办查询	待审批列表中包含新生成的审批编号	查询返回 <code>200</code> ，列表包含对应 <code>approval_id</code>	符合要求
医生批准	审批状态更新并创建订单任务	请求返回 <code>200</code> ， <code>order_created=true</code> ，生成 <code>task_id</code>	符合要求

比较项目	动态输出要求	实际动态输出	比较结果
库存变化	批准数量为 2 时，库存由 10 减少至 8	数据库实际库存为 8	符合要求
任务发布	发布函数接收正确的任务、药品和数量	Mock 捕获输出为 <code>(task_id, drug_id, 2)</code>	符合要求
流程完成	患者确认后审批状态变为 <code>completed</code>	完成请求返回 <code>200</code> ，最终状态为 <code>completed</code>	符合要求
事务原子性	批量订单任一项目无效时全部回滚	库存分别保持 10 和 1，订单、库存流水及发布调用数量均为 0	符合要求
直接订单审计	扣库、订单和库存流水数据一致	库存由 6 变为 4；订单数量为 2；流水变化量为 <code>-2</code>	符合要求

发现：

- 1. 患者、医生、库存、订单和任务发布调用之间的主业务链路能够完成，内部生成的审批编号和任务编号可在后续步骤继续使用。
- 2. 批量订单在库存不足时能够整体回滚，没有产生部分扣库或无效任务。
- 3. 直接订单能够生成库存流水，但审批自动下单未生成同类流水，因此两条下单路径的动态输出不一致。
- 4. 显式执行旧文件 `test-full-integration.py` 时得到 `2 passed`，但两个测试函数通过返回布尔值报告结果，`pytest` 同时输出 `PytestReturnNotNoneWarning`。该文件存在流程失败后仍被判为通过的风险，不能单独作为正式 E2E 验收依据。

3.4 测试4 (E2E-004: ROS2 集成测试)

本项测试使用 Mock/替身验证 ROS 配置、节点管理、消息转换、状态存储、健康监控、错误处理、任务发布和返回队列命令。

比较项目	动态输出要求	实际动态输出	比较结果
独立测试集执行	ROS2 集成用例成功率不低于 95%	141 passed in 15.82s , 成功率 100%	符合要求
消息格式转换	药品、货架、任务和小车状态正确转换	消息适配相关测试全部通过	符合要求
错误与降级处理	发布异常产生重试或降级结果，系统不崩溃	错误处理和健康监控测试通过	符合 Mock 测试要求

发现:

- 1. ROS2 Mock 测试已补齐 QoS、消息类型、发布者、订阅器和单例重置逻辑，原有非预期失败及顺序依赖未再出现。
- 2. 状态订阅回调、数据库状态更新、车辆状态和药柜状态均已增加测试；
state_subscriber.py 覆盖率由 17% 提升至 63%。
- 3. updated_at 尾缀 z 的生产代码兼容问题仍由严格 xfail 跟踪，记为 DEF-ROS-001。
- 4. 本项仅验证 Mock/替身环境，尚未取得真实 ROS2 话题、ROS-TCP-Endpoint、Unity 或机器人产生的动态输出。

3.5 测试5 (UT-002 ~ UT-005: Agent 及辅助模块测试)

本项测试覆盖 LLM 消息结构、MedicalAgent 基础行为、消息记忆、待办管理、工具注册执行和工作流状态管理。

比较项目	动态输出要求	实际动态输出	比较结果
用例执行	Agent 基础模块测试全部成功	143 passed in 8.84s	符合要求
消息与工具数据	序列化、反序列化和工具调用字段保持一致	LLM schema 和工具测试全部通过	符合要求
记忆管理	消息添加、重置和压缩后保留规定内容	记忆管理测试全部通过	符合要求

比较项目	动态输出要求	实际动态输出	比较结果
待办与工作流	状态变化、进度、统计和重置输出正确	待办及工作流测试全部通过	符合要求
异步调用	异步 Mock 应被正确等待	本次分组执行未产生协程未等待警告	符合要求

发现:

- 1. Agent 基础数据结构和内部状态输出与测试要求一致，未出现功能失败。
- 2. 新增表单状态机、症状规则/LLM 提取、药品匹配重试、危险信号、审批工具调用、会话持久化和规划器分支测试。
- 3. `medical_agent.py` 覆盖率达到 94%，`subagents/form.py` 达到 97%，`subagents/extractor.py` 达到 92%，消息管理器和待办模型均达到 98%。
- 4. LLM Client、OpenAI/Claude Provider、流式输出及网络异常分支已补充测试；`agent/llm/client.py` 覆盖率达到 98%。
- 5. 本项主要验证 Mock 环境中的行为，未验证真实 LLM 的医学准确率、响应时间和长对话稳定性。

3.6 测试6 (SYS-001: 全量回归与覆盖率测试)

本项测试比较整个自动化测试集的执行结果和生产代码覆盖率是否达到测试计划规定的系统准入要求。

比较项目	动态输出要求	实际动态输出	比较结果
全量回归	所有非预期测试成功执行	440 passed, 2 xfailed, 73 warnings in 91.24s	符合要求
非 xfail 通过率	不低于 95%	440 / 440 = 100%	达到比例要求
生产代码覆盖率	不低于 85%	85%	符合要求
HTML 报告	生成可查看的覆盖率 HTML 报告	生成 <code>coverage_report/index.html</code>	符合要求

发现:

- 1. 全量测试没有非预期失败，两个严格 xfail 均对应已登记缺陷。

2. 生产代码覆盖率达到 85%，满足系统整体覆盖率准入目标。
 3. MedicalAgent、表单状态机、症状提取、消息管理、待办模型、验证器及数据库初始化脚本的主要正常、异常和降级分支已补充测试。
 4. HTTP Client、订单控制器、配置服务、上传接口、数据库连接和症状数据库服务已达到 85% ~ 100% 覆盖率。
 5. 覆盖率 HTML 报告已重新生成，可按文件和缺失行继续定位剩余测试缺口。
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4 对软件功能的结论

4.1 认证与权限

注册、登录、刷新、注销、JWT 校验和角色权限测试通过，核心权限边界可用。但测试密钥长度不足，尚未执行令牌过期等待、密钥轮换、并发会话和渗透测试。

4.2 智能筛查与 Agent

症状标准化、同义词扩展、药品筛选、批量筛查、配置管理、历史记录，以及 Agent 的消息、记忆、工具、表单状态机和工作流能力通过测试。

MedicalAgent 覆盖率达到 94%，表单模块达到 97%，症状提取器达到 92%，筛查配置服务达到 96%，筛查路由达到 79%。

4.3 药品、库存、审批与订单

药品 CRUD、查询、分类、库存调整、文件上传和主要审批下单流程可执行，批量订单的事务原子性得到验证。订单控制器覆盖率达到 85%，但审批自动下单仍未生成库存流水。

4.4 ROS 状态与任务接口

配置、消息转换、状态订阅、任务发布、错误处理和健康监控的大部分逻辑通过 Mock 测试，ROS2 独立测试集 141 项全部通过。本次未验证真实 Unity 通信和硬件故障恢复。

5 分析摘要

5.1 能力

经测试，本软件具有如下能力：

1. Agent 的消息模型、记忆压缩、待办管理、工具注册和工作流状态管理等功能完善；
2. 用户注册、登录、令牌刷新与注销、角色权限、筛查历史和关键 API 权限边界等功能完善；

- 3. 药品 CRUD、搜索、分页、批量导入、导出、分类和库存调整等功能完善。
- 4. 患者提交审批、医生审批、扣减库存、创建任务和患者确认完成的主要业务等功能完善。
- 5. ROS 配置、错误处理、健康监控、消息适配、节点管理等功能完善。
- 6. HTTP 客户端、订单边界、上传接口、筛查配置和症状数据库服务已覆盖正常、异常和降级路径。

5.2 缺陷与限制汇总

编号	等级	缺陷或限制	建议
DEF-ROS-001	P2	<code>z</code> 时间戳解析失败, 在线状态误判	解析前将 <code>z</code> 转为 <code>+00:00</code> , 并补充时区测试
DEF-INT-001	P2	审批自动下单未写库存流水	复用统一订单事务逻辑, 保证库存、订单、流水原子提交
DEF-TEST-002	P3	E2E 文件未被默认发现且返回布尔值	重命名为 <code>test_full_integration.py</code> 并改用断言
DEF-SEC-001	P2	JWT 测试密钥长度不足	使用至少 32 字节的测试与部署密钥
DEF-DEP-001	P3	Screening ORM 使用 SQLAlchemy 2.0 已弃用的 <code>declarative_base()</code> 导入	改用 <code>sqlalchemy.orm.declarative_base</code> 或统一 <code>DeclarativeBase</code>
DEF-ENV-001	P3	尚未执行真实 ROS2、Unity 和机器人端到端测试	在目标部署环境补充真实话题与故障恢复验证

5.3 建议

系统的 Agent 表单与决策流程、认证授权、药品库存 API、订单、筛查配置和主要审批流程已具备继续开发与联调的基础。当前自动化测试已无非预期失败, 生产代码覆盖率已达到 85%, 但仍存在审批库存流水缺失、ROS 时间戳兼容及部分控制器覆盖偏低等问题。

修复后应在相同 Python 3.10 环境重跑全部 442 项默认测试及显式 E2E 测试，并在 Ubuntu ROS2 Humble、ROS-TCP-Endpoint 和 Unity 环境补做真实消息端到端验证。

5.4 评价

准入项	目标	实际	结论
全量测试	95% 成功	440 passed, 0 failed, 2 xfailed	通过
集成测试通过率	≥95%	14 passed, 2 个已知 xfail	通过
ROS2 集成成功率	≥85%	独立 Mock 测试 141/141, 100%	通过
生产代码覆盖率	≥80%	85%	通过
P1 缺陷	0	0	通过
P2 缺陷	≤5	3 项	通过
审计日志完整	完整	审批自动下单缺少库存流水	部分通过

该软件已达到自动化回归稳定、整体覆盖率达标和主要业务链路可联调的阶段。审计完整性及真实 ROS2/Unity 环境验证仍未完全满足正式交付要求，建议完成剩余 P2 问题并进行真实环境验证后再判定正式交付。

6 测试资源消耗

本次数据来自当前工作区自动化测试执行结果。机时消耗如下：

项目	实际消耗
默认全量测试	442 项，其中 440 passed、2 xfailed
覆盖率测试	442 项，91.24 秒
分组测试	unit、integration、ros_integration、agent、根目录 API/E2E
数据资源	测试夹具创建临时 SQLite 数据库

项目	实际消耗
报告产物	本 Markdown 报告; coverage_report/index.html ; report.html

Report generated on 07-Jun-2026 at 05:00:23 by [pytest-html](#) v4.2.0

Python	3.12.3
Platform	Linux-6.6.87.2-microsoft-standard-WSL2-x86_64-with-glibc2.39
Packages	• pytest: 9.0.3 • pluggy: 1.6.0
Plugins	• anyio: 4.13.0 • asyncio: 1.4.0 • cov: 7.1.0 • html: 4.2.0 • metadata: 3.1.1 • mock: 3.15.1

442 tests took 00:01:38.

(Un)check the boxes to filter the results.

2 Failed, 439 Passed, 0 Skipped, 1 Expected failures, 0 Unexpected passes, 0 Errors, 0 Reruns 0 Retried, [Show all details](#) / [Hide all details](#)

Passed	tests/agent/test_consultation_form_extended.py::test_models_serialize_and_rich_form_summarizes_all_sections
Passed	tests/agent/test_consultation_form_extended.py::test_basics_question_progression_and_field_access
Passed	tests/agent/test_consultation_form_extended.py::test_risk_condition_variants[patient.age < 12--True]
Passed	tests/agent/test_consultation_form_extended.py::test_risk_condition_variants[patient.age > 75--True]
Passed	tests/agent/test_consultation_form_extended.py::test_risk_condition_variants[gender_female--True]
Passed	tests/agent/test_consultation_form_extended.py::test_risk_condition_variants[chronic_diseases_not_empty--True]
Passed	tests/agent/test_consultation_form_extended.py::test_risk_condition_variants[social.need_operate_machinery--True]
Passed	tests/agent/test_consultation_form_extended.py::test_risk_condition_variants[unknown--False]
Passed	tests/agent/test_consultation_form_extended.py::test_completeness_handles_red_flags_missing_data_and_confirmation
Passed	tests/agent/test_consultation_form_extended.py::test_system_analysis_followups_and_form_summary
Passed	tests/agent/test_consultation_form_extended.py::test_incremental_extractor_merges_every_supported_section
Passed	tests/agent/test_consultation_form_extended.py::test_incremental_extractor_parsing_validation_and_fallback
Passed	tests/agent/test_consultation_form_extended.py::test_rule_extract_basics[patient.gender-\u7537-]
Passed	tests/agent/test_consultation_form_extended.py::test_rule_extract_basics[patient.gender-\u5973-]
Passed	tests/agent/test_consultation_form_extended.py::test_rule_extract_basics[patient.gender-\u4e0d\u613f\u900f\u9732-]
Passed	tests/agent/test_consultation_form_extended.py::test_rule_extract_basics[patient.age-18\u6708-]

Result 	Test
Passed	tests/agent/test_consultation_form_extended.py::test_rule_extract_basics[patient.weight_kg-140\u65a4-]
Passed	tests/agent/test_consultation_form_extended.py::test_rule_extract_basics[patient.weight_kg-65.5kg-]
Passed	tests/agent/test_consultation_form_extended.py::test_rule_extract_basics[allergies.drug_allergies-\u9752\u9709\u7d20\u3001\u78fa\u80fa-]
Passed	tests/agent/test_consultation_form_extended.py::test_rule_extract_basics[allergies.drug_allergies-\u65e0\u8fc7\u654f-]
Passed	tests/agent/test_consultation_form_extended.py::test_rule_extract_basics[current_medications-\u963fu53f8\u5339\u6797uff0c\u6c28\u6c2fu5730\u5e73-]
Passed	tests/agent/test_consultation_form_extended.py::test_rule_extract_basics[current_medications-\u6ca1\u6709\u7528\u836f-]
Passed	tests/agent/test_consultation_form_extended.py::test_rule_extract_basics_ignores_invalid_and_custom_values
Passed	tests/agent/test_llm_schemas.py::TestLLMMessage::test_create_user_message
Passed	tests/agent/test_llm_schemas.py::TestLLMMessage::test_create_assistant_with_tool_calls
Passed	tests/agent/test_llm_schemas.py::TestLLMMessage::test_to_dict
Passed	tests/agent/test_llm_schemas.py::TestLLMMessage::test_to_dict_with_tool_call_id
Passed	tests/agent/test_llm_schemas.py::TestLLMMessage::test_from_dict
Passed	tests/agent/test_llm_schemas.py::TestLLMMessage::test_from_dict_with_tool_calls
Passed	tests/agent/test_llm_schemas.py::TestToolCall::test_create_tool_call
Passed	tests/agent/test_llm_schemas.py::TestToolCall::test_to_dict
Passed	tests/agent/test_llm_schemas.py::TestToolCall::test_from_dict
Passed	tests/agent/test_llm_schemas.py::TestLLMResponse::test_create_response
Passed	tests/agent/test_llm_schemas.py::TestLLMResponse::test_to_dict
Passed	tests/agent/test_medical_agent.py::TestMedicalAgentInit::test_init_with_mocks
Passed	tests/agent/test_medical_agent.py::TestMedicalAgentInit::test_init_has_workflow_manager
Passed	tests/agent/test_medical_agent.py::TestMedicalAgentInit::test_init_tracking_fields
Passed	tests/agent/test_medical_agent.py::TestMedicalAgentReset::test_reset_clears_state
Passed	tests/agent/test_medical_agent.py::TestMedicalAgentRun::test_run_basic_with_mocks
Passed	tests/agent/test_medical_agent.py::TestMedicalAgentRun::test_run_returns_reply_content
Passed	tests/agent/test_medical_agent.py::TestMedicalAgentRun::test_run_tracks_patient_id
Passed	tests/agent/test_medical_agent.py::TestMedicalAgentSubmitBasics::test_submit_basics_exists
Passed	tests/agent/test_medical_agent.py::TestMedicalAgentGetApprovalStatus::test_get_approval_status_does_not_crash
Passed	tests/agent/test_medical_agent_extended.py::test_submit_basics_accepts_units_strings_lists_and_invalid_numbers
Passed	tests/agent/test_medical_agent_extended.py::test_form_session_round_trip_and_invalid_drug_matching_state
Passed	tests/agent/test_medical_agent_extended.py::test_confirmation_keywords_and_initial_basics_question
Passed	tests/agent/test_medical_agent_extended.py::test_basics_answer_can_finish_round_one
Passed	tests/agent/test_medical_agent_extended.py::test_form_collection_handles_red_flag_continue_and_confirmation
Passed	tests/agent/test_medical_agent_extended.py::test_confirmation_phase_rejects_empty_symptoms_accepts_valid_form_and_edits
Passed	tests/agent/test_medical_agent_extended.py::test_query_drugs_uses_direct_matches_name_lookup_and_llm_synonyms
Passed	tests/agent/test_medical_agent_extended.py::test_confirmed_form_builds_full_context_and_returns_llm_reply
Passed	tests/agent/test_medical_agent_extended.py::test_confirmed_form_rejects_empty_symptoms_red_flags_and_no_drugs
Passed	tests/agent/test_medical_agent_extended.py::test_legacy_path_handles_red_flags_no_drugs_and_tool_loop
Passed	tests/agent/test_medical_agent_extended.py::test_cached_query_tool_and_tool_error_paths
Passed	tests/agent/test_medical_agent_extended.py::test_workflow_accessors_approval_status_and_state_persistence
Passed	tests/agent/test_medical_agent_extended.py::test_planner_plan_formatting_and_todo_completion
Passed	tests/agent/test_medical_agent_extended.py::test_run_form_dispatch_and_remaining_form_phase_branches
Passed	tests/agent/test_medical_agent_extended.py::test_query_retry_rejects_irrelevant_matches_and_survives_llm_errors
Passed	tests/agent/test_medical_agent_extended.py::test_tool_error_patient_name_reset_and_approval_failure_paths
Passed	tests/agent/test_medical_agent_extended.py::test_load_state_restores_tool_messages
Passed	tests/agent/test_memory.py::TestMessageManagerInit::test_init_with_system_prompt
Passed	tests/agent/test_memory.py::TestMessageManagerInit::test_init_without_system_prompt
Passed	tests/agent/test_memory.py::TestMessageManagerInit::test_get_messages_without_system
Passed	tests/agent/test_memory.py::TestMessageManagerAdd::test_add_user_message
Passed	tests/agent/test_memory.py::TestMessageManagerAdd::test_add_assistant_message
Passed	tests/agent/test_memory.py::TestMessageManagerAdd::test_add_tool_result
Passed	tests/agent/test_memory.py::TestMessageManagerAdd::test_add_tool_message_with_dict_content
Passed	tests/agent/test_memory.py::TestMessageManagerReset::test_reset_keeps_system
Passed	tests/agent/test_memory.py::TestMessageManagerReset::test_reset_without_system
Passed	tests/agent/test_memory.py::TestCompressByTokens::test_basic_compression
Passed	tests/agent/test_memory.py::TestCompressByTokens::test_compress_does_not_remove_system
Passed	tests/agent/test_memory.py::TestCompressByTokens::test_compress_removes_all_when_no_preserve
Passed	tests/agent/test_memory.py::TestCompressByCount::test_count_compression
Passed	tests/agent/test_memory.py::TestCompressByCount::test_count_preserves_system
Passed	tests/agent/test_memory.py::TestCompressByCount::test_count_noop_when_under_limit
Passed	tests/agent/test_memory.py::TestSmartCompress::test_smart_compress
Passed	tests/agent/test_symptom_extractor_extended.py::test_rule_extraction_collects_patient_symptoms_signs_and_history

Result 	Test
Passed	tests/agent/test_symptom_extractor_extended.py::test_rule_extraction_honors_negation_and_extracts_alternate_sign_formats
Passed	tests/agent/test_symptom_extractor_extended.py::test_llm_extraction_cleans_degree_words_and_parses_embedded_json
Passed	tests/agent/test_symptom_extractor_extended.py::test_llm_failure_and_missing_client_fall_back_to_rules
Passed	tests/agent/test_symptom_extractor_extended.py::test_async_extraction_supports_rule_and_llm_modes
Passed	tests/agent/test_symptom_extractor_extended.py::test_empty_input_is_rejected[]
Passed	tests/agent/test_symptom_extractor_extended.py::test_empty_input_is_rejected[]
Passed	tests/agent/test_symptom_extractor_extended.py::test_empty_input_is_rejected[None]
Passed	tests/agent/test_symptom_extractor_extended.py::test_validation_reports_each_invalid_shape
Passed	tests/agent/test_symptom_extractor_extended.py::test_helpers_cover_cleaning_prompt_and_json_errors
Passed	tests/agent/test_todo.py::TestTaskStatus::test_status_values
Passed	tests/agent/test_todo.py::TestTaskCategory::test_category_values
Passed	tests/agent/test_todo.py::TestTodoTask::test_default_creation
Passed	tests/agent/test_todo.py::TestTodoTask::test_custom_creation
Passed	tests/agent/test_todo.py::TestTodoTask::test_mark_completed
Passed	tests/agent/test_todo.py::TestTodoTask::test_mark_in_progress
Passed	tests/agent/test_todo.py::TestTodoTask::test_mark_blocked
Passed	tests/agent/test_todo.py::TestTodoTask::test_is_completed
Passed	tests/agent/test_todo.py::TestTodoTask::test_get_age
Passed	tests/agent/test_todo.py::TestTodoTask::test_to_dict
Passed	tests/agent/test_todo.py::TestTodoTask::test_from_dict
Passed	tests/agent/test_todo.py::TestTodoManager::test_add_and_get_todo
Passed	tests/agent/test_todo.py::TestTodoManager::test_get_nonexistent_returns_none
Passed	tests/agent/test_todo.py::TestTodoManager::test_get_todo_list
Passed	tests/agent/test_todo.py::TestTodoManager::test_get_pending_tasks
Passed	tests/agent/test_todo.py::TestTodoManager::test_get_high_priority_tasks
Passed	tests/agent/test_todo.py::TestTodoManager::test_get_tasks_by_category
Passed	tests/agent/test_todo.py::TestTodoManager::test_update_todo
Passed	tests/agent/test_todo.py::TestTodoManager::test_manager_mark_completed
Passed	tests/agent/test_todo.py::TestTodoManager::test_manager_mark_in_progress
Passed	tests/agent/test_todo.py::TestTodoManager::test_delete_todo
Passed	tests/agent/test_todo.py::TestTodoManager::test_clear_completed_tasks
Passed	tests/agent/test_todo.py::TestTodoManager::test_get_summary
Passed	tests/agent/test_tools.py::TestToolExecutor::test_register_and_execute
Passed	tests/agent/test_tools.py::TestToolExecutor::test_execute_unregistered_raises
Passed	tests/agent/test_tools.py::TestToolExecutor::test_execute_handler_error
Passed	tests/agent/test_tools.py::TestBaseTool::test_cannot_instantiate_abstract
Passed	tests/agent/test_tools.py::TestBaseTool::test_concrete_tool
Passed	tests/agent/test_tools.py::TestBaseTool::test_validate_input
Passed	tests/agent/test_tools.py::TestToolSchemas::test_tools_list_has_six_tools
Passed	tests/agent/test_tools.py::TestToolSchemas::test_each_tool_has_required_keys
Passed	tests/agent/test_tools.py::TestToolSchemas::test_query_drug_schema
Passed	tests/agent/test_tools.py::TestToolSchemas::test_submit_approval_schema
Passed	tests/agent/test_tools.py::TestToolSchemas::test_check_allergy_schema
Passed	tests/agent/test_tools.py::TestToolSchemas::test_calc_dosage_schema
Passed	tests/agent/test_tools.py::TestToolSchemas::test_generate_advice_schema
Passed	tests/agent/test_tools.py::TestToolSchemas::test_fill_prescription_schema
Passed	tests/agent/test_tools.py::TestRegistryFunctions::test_execute_tool_function
Passed	tests/agent/test_tools.py::TestRegistryFunctions::test_get_registered_tools
Passed	tests/agent/test_workflow.py::TestWorkflowStep::test_step_values
Passed	tests/agent/test_workflow.py::TestWorkflowStep::test_step_count
Passed	tests/agent/test_workflow.py::TestWorkflowState::test_initial_state
Passed	tests/agent/test_workflow.py::TestWorkflowState::test_mark_step_completed
Passed	tests/agent/test_workflow.py::TestWorkflowState::test_multiple_steps
Passed	tests/agent/test_workflow.py::TestWorkflowState::test_get_progress
Passed	tests/agent/test_workflow.py::TestWorkflowState::test_get_duration
Passed	tests/agent/test_workflow.py::TestWorkflowState::test_to_dict
Passed	tests/agent/test_workflow.py::TestWorkflowManager::test_create_and_get_workflow
Passed	tests/agent/test_workflow.py::TestWorkflowManager::test_get_nonexistent_returns_none
Passed	tests/agent/test_workflow.py::TestWorkflowManager::test_update_workflow_step
Passed	tests/agent/test_workflow.py::TestWorkflowManager::test_update_nonexistent_creates_workflow
Passed	tests/agent/test_workflow.py::TestWorkflowManager::test_set_approval_id

Result 	Test
Passed	tests/agent/test_workflow.py::TestWorkflowManager::test_get_all_workflows
Passed	tests/agent/test_workflow.py::TestWorkflowManager::test_clear_completed
Passed	tests/agent/test_workflow.py::TestWorkflowManager::test_reset_workflow
Passed	tests/agent/test_workflow.py::TestWorkflowManager::test_get_stats_empty
Passed	tests/agent/test_workflow.py::TestWorkflowManager::test_get_stats
Passed	tests/integration/test_auth_screening.py::test_auth_registration_refresh_logout_lifecycle
Passed	tests/integration/test_auth_screening.py::test_patient_screening_and_history_flow
Passed	tests/integration/test_auth_screening.py::test_role_permissions_span_drug_and_approval_apis
Passed	tests/integration/test_medication_workflow.py::test_patient_approval_doctor_dispatch_and_completion
Passed	tests/integration/test_medication_workflow.py::test_batch_order_is_atomic_when_any_item_is_invalid
Passed	tests/integration/test_medication_workflow.py::test_direct_order_writes_inventory_audit_and_order_record
Passed	tests/integration/test_order_controller_edges.py::test_order_and_pickup_validation_errors
Passed	tests/integration/test_order_controller_edges.py::test_pickup_list_complete_and_not_found
Passed	tests/integration/test_order_controller_edges.py::test_dispense_validation_success_and_atomic_failure
Passed	tests/integration/test_order_controller_edges.py::test_create_order_for_drug_missing_and_insufficient
Passed	tests/integration/test_ros_state_api.py::test_ros_subscription_state_is_exposed_through_http
Passed	tests/integration/test_ros_state_api.py::test_ros_return_command_requires_inventory_permission
Passed	tests/integration/test_screening_routes_edges.py::test_screening_symptom_batch_config_and_status_edges
Passed	tests/integration/test_screening_routes_edges.py::test_screening_history_validation_and_detail
Passed	tests/ros_integration/test_bridge.py::TestPublishTask::test_publish_task_lazy_initializes_publisher
Passed	tests/ros_integration/test_bridge.py::TestPublishTask::test_publish_task_calls_publish
Passed	tests/ros_integration/test_bridge.py::TestPublishTask::test_publish_task_handles_exception
Passed	tests/ros_integration/test_bridge.py::TestPublishTask::test_publish_task_unavailable_and_failed_result
Passed	tests/ros_integration/test_bridge.py::TestPublishExpiryRemoval::test_publish_expiry_removal_calls_publish
Passed	tests/ros_integration/test_bridge.py::TestPublishExpiryRemoval::test_publish_expiry_removal_lazy_init
Passed	tests/ros_integration/test_bridge.py::TestPublishExpiryRemoval::test_publish_expiry_failure_and_exception
Passed	tests/ros_integration/test_bridge.py::TestPublishReturnToQueue::test_publish_return_to_queue_calls_publish
Passed	tests/ros_integration/test_bridge.py::TestPublishReturnToQueue::test_return_failure_exception_and_unavailable
Passed	tests/ros_integration/test_bridge.py::TestCheckRos2Status::test_check_ros2_status_returns_dict
Passed	tests/ros_integration/test_bridge.py::TestCheckRos2Status::test_check_ros2_status_legacy_fallback
Passed	tests/ros_integration/test_bridge.py::TestCheckRos2Status::test_check_ros2_status_with_node_running
Passed	tests/ros_integration/test_bridge.py::TestCheckRos2Status::test_check_ros2_status_error
Passed	tests/ros_integration/test_config.py::TestConfigDefaults::test_default_integration_mode
Passed	tests/ros_integration/test_config.py::TestConfigDefaults::test_default_ros_domain_id
Passed	tests/ros_integration/test_config.py::TestConfigDefaults::test_default_tcp_settings
Passed	tests/ros_integration/test_config.py::TestConfigDefaults::test_default_topic_names
Passed	tests/ros_integration/test_config.py::TestConfigDefaults::test_default_performance_settings
Passed	tests/ros_integration/test_config.py::TestConfigDefaults::test_default_health_settings
Passed	tests/ros_integration/test_config.py::TestConfigDefaults::test_ros_node_name
Passed	tests/ros_integration/test_config.py::TestConfigEnvOverrides::test_integration_mode_from_env
Passed	tests/ros_integration/test_config.py::TestConfigEnvOverrides::test_ros_domain_id_from_env
Passed	tests/ros_integration/test_config.py::TestConfigEnvOverrides::test_tcp_port_from_env
Passed	tests/ros_integration/test_config.py::TestConfigEnvOverrides::test_tcp_ip_from_env
Passed	tests/ros_integration/test_config.py::TestConfigEnvOverrides::test_tcp_disabled_from_env
Passed	tests/ros_integration/test_config.py::TestConfigValidation::test_validate_valid
Passed	tests/ros_integration/test_config.py::TestConfigValidation::test_validate_invalid_mode
Passed	tests/ros_integration/test_config.py::TestConfigValidation::test_validate_invalid_tcp_port_zero
Passed	tests/ros_integration/test_config.py::TestConfigValidation::test_validate_invalid_tcp_port_negative
Passed	tests/ros_integration/test_config.py::TestConfigValidation::test_validate_invalid_tcp_port_overflow
Passed	tests/ros_integration/test_config.py::TestTopicConfig::test_topic_constants
Passed	tests/ros_integration/test_config.py::TestTopicConfig::test_qos_settings
Passed	tests/ros_integration/test_error_handler.py::TestErrorLevel::test_error_level_values
Passed	tests/ros_integration/test_error_handler.py::TestErrorLevel::test_error_handler_initial_stats
Passed	tests/ros_integration/test_error_handler.py::TestHandlePublishError::test_timeout_error_returns_wait_and_retry
Passed	tests/ros_integration/test_error_handler.py::TestHandlePublishError::test_timeout_error_linear_strategy
Passed	tests/ros_integration/test_error_handler.py::TestHandlePublishError::test_timeout_error_none_strategy
Passed	tests/ros_integration/test_error_handler.py::TestHandlePublishError::test_connection_error
Passed	tests/ros_integration/test_error_handler.py::TestHandlePublishError::test_unknown_error_requires_intervention
Passed	tests/ros_integration/test_error_handler.py::TestHandlePublishError::test_ros_connection_error_attempts_recovery
Passed	tests/ros_integration/test_error_handler.py::TestExponentialBackoff::test_backoff_depends_on_task_id
Passed	tests/ros_integration/test_error_handler.py::TestExponentialBackoff::test_backoff_capped_at_300

Result 	Test
Passed	tests/ros_integration/test_error_handler.py::TestErrorStats::test_stats_after_errors
Passed	tests/ros_integration/test_error_handler.py::TestErrorStats::test_reset_stats
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationInitial::test_initial_mode_is_full
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationInitial::test_initial_can_publish
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationInitial::test_initial_can_receive
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationInitial::test_initial_is_fully_functional
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationInitial::test_initial_mode_name
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationThresholds::test_degrade_after_5_failures
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationThresholds::test_degrade_after_15_failures
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationThresholds::test_degrade_after_30_failures
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationThresholds::test_cannot_publish_in_log_only
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationThresholds::test_cannot_receive_in_publish_only
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationThresholds::test_cannot_publish_in_disabled
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationRecovery::test_recovery_after_success
Passed	tests/ros_integration/test_error_handler.py::TestGracefulDegradationRecovery::test_reset_restores_full
Passed	tests/ros_integration/test_error_handler.py::TestDegradationModeEnum::test_mode_ordering
Passed	tests/ros_integration/test_health_monitor.py::TestHealthMonitorInit::test_initial_state
Passed	tests/ros_integration/test_health_monitor.py::TestHealthMonitorInit::test_initial_get_health_status
Passed	tests/ros_integration/test_health_monitor.py::TestHealthMonitorInit::test_initial_is_healthy_false
Passed	tests/ros_integration/test_health_monitor.py::TestStartStop::test_start_creates_monitor_thread
Passed	tests/ros_integration/test_health_monitor.py::TestStartStop::test_double_start_is_noop
Passed	tests/ros_integration/test_health_monitor.py::TestStartStop::test_stop_joins_thread
Passed	tests/ros_integration/test_health_monitor.py::TestStartStop::test_stop_when_not_started
Passed	tests/ros_integration/test_health_monitor.py::TestPerformHealthCheck::test_perform_check_updates_timestamp
Passed	tests/ros_integration/test_health_monitor.py::TestPerformHealthCheck::test_perform_check_updates_connection_healthy
Passed	tests/ros_integration/test_health_monitor.py::TestPerformHealthCheck::test_perform_check_with_connection_failure
Passed	tests/ros_integration/test_health_monitor.py::TestPerformHealthCheck::test_perform_check_increments_consecutive_failures
Passed	tests/ros_integration/test_health_monitor.py::TestPerformHealthCheck::test_perform_check_decreases_consecutive_failures_on_success
Passed	tests/ros_integration/test_health_monitor.py::TestIsHealthy::test_is_healthy_true
Passed	tests/ros_integration/test_health_monitor.py::TestIsHealthy::test_is_healthy_false_when_not_connected
Passed	tests/ros_integration/test_health_monitor.py::TestIsHealthy::test_is_healthy_false_when_high_error_rate
Passed	tests/ros_integration/test_health_monitor.py::TestCheckDegradation::test_degradation_failure_on_connection_lost
Passed	tests/ros_integration/test_health_monitor.py::TestCheckDegradation::test_degradation_success_on_connection_good
Passed	tests/ros_integration/test_health_monitor.py::TestReset::test_reset_clears_state
Passed	tests/ros_integration/test_message_adapter.py::TestSingleton::test_singleton
Passed	tests/ros_integration/test_message_adapter.py::TestBackendToUnity::test_valid_drug_returns_task_msg
Passed	tests/ros_integration/test_message_adapter.py::TestBackendToUnity::test_shelf_mapping
Passed	tests/ros_integration/test_message_adapter.py::TestBackendToUnity::test_missing_shelve_id_raises_key_error
Passed	tests/ros_integration/test_message_adapter.py::TestBackendToUnity::test_missing_shelf_x_raises_key_error
Passed	tests/ros_integration/test_message_adapter.py::TestBackendToUnity::test_invalid_quantity_zero_raises_value_error
Passed	tests/ros_integration/test_message_adapter.py::TestBackendToUnity::test_quantity_exceeds_stock_raises_value_error
Passed	tests/ros_integration/test_message_adapter.py::TestExpiryToUnity::test_expiry_removal_returns_task_msg
Passed	tests/ros_integration/test_message_adapter.py::TestExpiryToUnity::test_expiry_missing_field_raises_key_error
Passed	tests/ros_integration/test_message_adapter.py::TestUnityToBackend::test_converts_task_state
Passed	tests/ros_integration/test_message_adapter.py::TestUnityToBackend::test_converts_without_car_id
Passed	tests/ros_integration/test_message_adapter.py::TestValidateMapping::test_valid_mapping
Passed	tests/ros_integration/test_message_adapter.py::TestValidateMapping::test_invalid_mapping_out_of_range
Passed	tests/ros_integration/test_message_adapter.py::TestValidateMapping::test_invalid_mapping_missing_fields
Passed	tests/ros_integration/test_message_adapter.py::TestValidateMapping::test_invalid_mapping_non_numeric
Passed	tests/ros_integration/test_node_manager.py::TestSingleton::test_get_instance_returns_same
Passed	tests/ros_integration/test_node_manager.py::TestSingleton::test_get_instance_creates_on_demand
Passed	tests/ros_integration/test_node_manager.py::TestInitRos2::test_init_creates_node_and_executor
Passed	tests/ros_integration/test_node_manager.py::TestInitRos2::test_init_registers_atexit
Passed	tests/ros_integration/test_node_manager.py::TestInitRos2::test_atexit_callback_does_not_keep_discarded_instance_alive
Passed	tests/ros_integration/test_node_manager.py::TestCheckConnection::test_check_connection_returns_true_when_ok
Passed	tests/ros_integration/test_node_manager.py::TestCheckConnection::test_check_connection_false_when_rclpy_not_ok
Passed	tests/ros_integration/test_node_manager.py::TestStartShutdown::test_start_creates_executor_thread
Passed	tests/ros_integration/test_node_manager.py::TestStartShutdown::test_shutdown_stops_executor_thread
Passed	tests/ros_integration/test_node_manager.py::TestStartShutdown::test_atexit_shutdown_handles_invalid_test_doubles
Passed	tests/ros_integration/test_node_manager.py::TestStartShutdown::test_is_running_false_when_no_thread
Passed	tests/ros_integration/test_node_manager.py::TestStartShutdown::test_is_running_true_when_alive

Result 	Test
Passed	tests/ros_integration/test_node_manager.py::TestGetNodeAndExecutor::test_get_node_returns_node
Passed	tests/ros_integration/test_node_manager.py::TestGetNodeAndExecutor::test_get_executor_returns_executor
Passed	tests/ros_integration/test_node_manager.py::TestOptimalThreadCount::test_get_optimal_thread_count
Passed	tests/ros_integration/test_state_store.py::TestSingleton::test_singleton_returns_same_instance
Passed	tests/ros_integration/test_state_store.py::TestSingleton::test_singleton_initialized_once
Passed	tests/ros_integration/test_state_store.py::TestSingleton::test_reset_singleton
Passed	tests/ros_integration/test_state_store.py::TestCarState::test_update_and_get_car
Passed	tests/ros_integration/test_state_store.py::TestCarState::test_update_multiple_cars
Passed	tests/ros_integration/test_state_store.py::TestCarState::test_update_same_car_overwrites
Passed	tests/ros_integration/test_state_store.py::TestTaskState::test_update_and_get_task
Passed	tests/ros_integration/test_state_store.py::TestTaskState::test_get_single_task
Passed	tests/ros_integration/test_state_store.py::TestTaskState::test_get_nonexistent_task
Passed	tests/ros_integration/test_state_store.py::TestTaskState::test_update_multiple_tasks
Passed	tests/ros_integration/test_state_store.py::TestCabinetState::test_update_and_get_cabinet
Passed	tests/ros_integration/test_state_store.py::TestCabinetState::test_update_multiple_cabinets
Passed	tests/ros_integration/test_state_store.py::TestGetAll::test_get_all_returns_all_states
Passed	tests/ros_integration/test_state_store.py::TestIsConnected::test_not_connected_when_empty
Passed	tests/ros_integration/test_state_store.py::TestIsConnected::test_connected_with_recent_update
Passed	tests/ros_integration/test_state_store.py::TestIsConnected::test_connected_with_recent_task_update
Passed	tests/ros_integration/test_state_store.py::TestIsConnected::test_invalid_timestamp_is_ignored
Passed	tests/ros_integration/test_state_store.py::TestThreadSafety::test_concurrent_updates
Passed	tests/ros_integration/test_state_subscriber.py::test_callback_setters_start_and_state_updates
Passed	tests/ros_integration/test_state_subscriber.py::test_task_callback_error_is_reported
Passed	tests/ros_integration/test_state_subscriber.py::test_user_callback_errors_are_swallowed
Passed	tests/ros_integration/test_task_publisher.py::TestInitPublishers::test_default_new_mode
Passed	tests/ros_integration/test_task_publisher.py::TestInitPublishers::test_legacy_mode
Passed	tests/ros_integration/test_task_publisher.py::TestInitPublishers::test_parallel_mode
Passed	tests/ros_integration/test_task_publisher.py::TestPublishTask::test_publish_task_new_success
Passed	tests/ros_integration/test_task_publisher.py::TestPublishTask::test_publish_task_new_success_called_with_msg
Passed	tests/ros_integration/test_task_publisher.py::TestPublishTask::test_publish_task_legacy_success
Passed	tests/ros_integration/test_task_publisher.py::TestPublishTask::test_publish_fails_when_not_initialized
Passed	tests/ros_integration/test_task_publisher.py::TestPublishTask::test_publish_fails_when_degraded
Passed	tests/ros_integration/test_task_publisher.py::TestPublishExpiryRemoval::test_expiry_removal_success_new
Passed	tests/ros_integration/test_task_publisher.py::TestPublishExpiryRemoval::test_expiry_removal_fails_when_not_initialized
Passed	tests/ros_integration/test_task_publisher.py::TestPublishReturnToQueue::test_return_to_queue_success_new
Passed	tests/ros_integration/test_task_publisher.py::TestPublishReturnToQueue::test_return_to_queue_fails_when_not_initialized
Passed	tests/ros_integration/test_task_publisher.py::TestDegradationInteraction::test_success_records_degradation_success
Passed	tests/ros_integration/test_task_publisher.py::TestDegradationInteraction::test_failure_records_degradation_failure
Passed	tests/test_backend_config.py::test_config_class_exists
Passed	tests/test_backend_config.py::test_config_default_values
Passed	tests/test_backend_config.py::test_config_port_validation
Passed	tests/test_backend_config.py::test_config_environment_overrides
Passed	tests/test_backend_config.py::test_config_to_dict_method
Passed	tests/test_backend_config.py::test_config_ros2_boolean_parsing
Passed	tests/test_backend_config.py::test_config_database_path_default_is_absolute
Passed	tests/test_drug_api.py::TestDrugAuth::test_list_drugs_requires_token
Passed	tests/test_drug_api.py::TestDrugAuth::test_patient_cannot_delete_drug
Passed	tests/test_drug_api.py::TestDrugCrud::test_create_validation_error
Passed	tests/test_drug_api.py::TestDrugCrud::test_create_get_update_delete
Passed	tests/test_drug_api.py::TestDrugListAndSearch::test_list_with_pagination_and_symptom
Passed	tests/test_drug_api.py::TestDrugListAndSearch::test_search_by_keyword
Passed	tests/test_drug_api.py::TestDrugListAndSearch::test_search_without_keyword_like_list
Passed	tests/test_drug_api.py::TestDrugBatchAndExport::test_batch_import
Passed	tests/test_drug_api.py::TestDrugBatchAndExport::test_batch_import_validation_stops_all
Passed	tests/test_drug_api.py::TestDrugBatchAndExport::test_export_json
Passed	tests/test_drug_api.py::TestDrugBatchAndExport::test_export_csv
Passed	tests/test_drug_api.py::TestCategories::test_list_and_create_category
Passed	tests/test_drug_api.py::TestCategories::test_category_tree
Passed	tests/test_inventory_api.py::TestInventoryAccess::test_patient_forbidden_inventory
Passed	tests/test_inventory_api.py::TestInventoryView::test_inventory_projection_fields
Passed	tests/test_inventory_api.py::TestAdjustInventory::test_adjust_out_insufficient_stock

Result 	Test
Passed	tests/test_inventory_api.py::TestInventoryAlerts::test_low_stock_list
Passed	tests/test_inventory_api.py::TestInventoryAlerts::test_low_stock_respects_min_stock_alert
Passed	tests/test_inventory_api.py::TestInventoryAlerts::test_expiring_soon_list
Passed	tests/test_inventory_api.py::TestInventoryAlerts::test_drug_stats
Passed	tests/unit/test_agent_support_services.py::test_symptom_service_rejects_blank_input
Passed	tests/unit/test_agent_support_services.py::test_symptom_service_selects_rule_and_llm_modes
Passed	tests/unit/test_agent_support_services.py::test_symptom_service_correction_combines_inputs
Passed	tests/unit/test_agent_support_services.py::test_record_transaction_outbound[order_result0-True]
Passed	tests/unit/test_agent_support_services.py::test_record_transaction_outbound[order_result1-False]
Passed	tests/unit/test_agent_support_services.py::test_record_transaction_outbound[None-False]
Passed	tests/unit/test_agent_support_services.py::test_record_transaction_inbound_and_exception
Passed	tests/unit/test_agent_support_services.py::test_stock_report_uses_backend_data
Passed	tests/unit/test_agent_support_services.py::test_stock_report_falls_back_when_backend_fails
Passed	tests/unit/test_agent_support_services.py::test_inventory_placeholder_outputs_are_structured_json
Passed	tests/unit/test_agent_support_services.py::test_report_generators_accept_explicit_periods
Passed	tests/unit/test_auth.py::test_access_token_round_trip
Passed	tests/unit/test_auth.py::test_invalid_access_token_is_rejected
Passed	tests/unit/test_auth.py::test_refresh_helpers_return_unique_future_values
Passed	tests/unit/test_auth.py::test_get_bearer_token_and_current_user
Passed	tests/unit/test_auth.py::test_non_access_or_malformed_subject_is_rejected
Passed	tests/unit/test_auth.py::test_auth_schema_seeds_roles_permissions_and_view
Passed	tests/unit/test_auth.py::test_require_auth_returns_401_without_token
Passed	tests/unit/test_auth.py::test_user_has_any_permission[perms0-needed0-True]
Passed	tests/unit/test_auth.py::test_user_has_any_permission[perms1-needed1-False]
Passed	tests/unit/test_auth.py::test_user_has_any_permission[perms2-needed2-False]
Passed	tests/unit/test_common_utils.py::test_validate_drug_accepts_valid_create_payload
Passed	tests/unit/test_common_utils.py::test_validate_drug_reports_multiple_invalid_fields
Passed	tests/unit/test_common_utils.py::test_validate_drug_update_allows_partial_payload
Passed	tests/unit/test_common_utils.py::test_validation_helpers_accept_valid_values[validate_category-payload0]
Passed	tests/unit/test_common_utils.py::test_validation_helpers_accept_valid_values[validate_inventory_transaction-payload1]
Passed	tests/unit/test_common_utils.py::test_validation_helpers_accept_valid_values[validate_pagination_params-payload2]
Passed	tests/unit/test_common_utils.py::test_validation_helpers_reject_invalid_values
Passed	tests/unit/test_common_utils.py::test_json_helpers_extract_parse_and_format
Passed	tests/unit/test_common_utils.py::test_text_helpers_cover_nested_access_merge_and_summary
Passed	tests/unit/test_common_utils.py::test_simple_cache_expiration_and_deletion
Passed	tests/unit/test_common_utils.py::test_drug_cache_invalidation
Passed	tests/unit/test_common_utils.py::test_retry_retries_then_returns
Passed	tests/unit/test_common_utils.py::test_retry_reraises_last_error
Passed	tests/unit/test_common_utils.py::test_pagination_and_exception_messages
Passed	tests/unit/test_config_service_extended.py::test_config_service_loads_valid_and_ignores_invalid_rows
Passed	tests/unit/test_config_service_extended.py::test_config_service_create_update_delete_round_trip
Passed	tests/unit/test_config_service_extended.py::test_config_service_database_failures
Passed	tests/unit/test_coverage_small_modules.py::test_drug_validation_covers_optional_types_choices_and_json_columns
Passed	tests/unit/test_coverage_small_modules.py::test_drug_validation_handles_missing_fields_long_indications_and_valid_options
Passed	tests/unit/test_coverage_small_modules.py::test_other_validators_report_type_and_length_errors[validate_category-payload0-fields0]
Passed	tests/unit/test_coverage_small_modules.py::test_other_validators_report_type_and_length_errors[validate_inventory_transaction-payload1-fields1]
Passed	tests/unit/test_coverage_small_modules.py::test_other_validators_report_type_and_length_errors[validate_pagination_params-payload2-fields2]
Passed	tests/unit/test_coverage_small_modules.py::test_message_manager_metadata_queries_and_both_compression_paths
Passed	tests/unit/test_coverage_small_modules.py::test_todo_edge_validation_storage_and_state_transitions
Passed	tests/unit/test_coverage_small_modules.py::test_todo_sort_filter_clear_and_no_storage_loader
Passed	tests/unit/test_coverage_small_modules.py::test_claude_provider_maps_messages_tools_usage_and_errors
Passed	tests/unit/test_coverage_small_modules.py::test_openai_provider_maps_native_and_embedded_tool_calls_and_errors
Passed	tests/unit/test_coverage_small_modules.py::test_logger_text_response_and_engine_convenience_wrappers
Passed	tests/unit/test_coverage_small_modules.py::test_backend_init_db_creates_seed_rows
Passed	tests/unit/test_database_helpers_and_screening_models.py::test_database_initialization_and_helper_queries
Passed	tests/unit/test_database_helpers_and_screening_models.py::test_database_migration_helpers_and_serializer
Passed	tests/unit/test_database_helpers_and_screening_models.py::test_screening_models_serialize_values
Passed	tests/unit/test_database_services.py::test_approval_manager_full_state_transition
Passed	tests/unit/test_database_services.py::test_approval_manager_rejects_only_pending_requests
Passed	tests/unit/test_database_services.py::test_pharmacy_client_query_fallbacks
Passed	tests/unit/test_database_services.py::test_pharmacy_client_update_stock_routes_operations

Result 	Test
Passed	tests/unit/test_database_services.py::test_pharmacy_client_converts_transport_errors_to_safe_defaults
Passed	tests/unit/test_database_services.py::test_pharmacy_client_wrapper_methods
Passed	tests/unit/test_database_services.py::test_pharmacy_client_wrapper_error_defaults
Passed	tests/unit/test_drug_controller.py::TestDrugController::test_create_drug
Passed	tests/unit/test_drug_controller.py::TestDrugController::test_delete_drug
Passed	tests/unit/test_drug_controller.py::TestDrugController::test_adjust_inventory
Passed	tests/unit/test_drug_controller.py::TestDrugController::test_query_drug
Passed	tests/unit/test_drug_controller.py::TestDrugController::test_low_stock
Passed	tests/unit/test_drug_controller.py::TestDrugController::test_expiring_soon
Passed	tests/unit/test_http_client_extended.py::test_client_validation_headers_extractors_and_async_runner
Passed	tests/unit/test_http_client_extended.py::test_make_request_success_http_and_network_errors
Passed	tests/unit/test_http_client_extended.py::test_make_raw_request_success_and_failure
Passed	tests/unit/test_http_client_extended.py::test_drug_inventory_and_category_async_methods
Passed	tests/unit/test_http_client_extended.py::test_export_csv_and_approval_order_health_methods
Passed	tests/unit/test_http_client_extended.py::test_sync_wrappers_delegate_to_run_async
Passed	tests/unit/test_llm_client_and_providers.py::test_llm_client_rejects_unknown_provider
Passed	tests/unit/test_llm_client_and_providers.py::test_llm_client_chat_updates_stats
Passed	tests/unit/test_llm_client_and_providers.py::test_llm_client_estimates_tokens_without_usage
Passed	tests/unit/test_llm_client_and_providers.py::test_openai_stream_yields_nonempty_chunks
Passed	tests/unit/test_llm_client_and_providers.py::test_non_openai_stream_uses_regular_chat
Passed	tests/unit/test_llm_client_and_providers.py::test_openai_provider_maps_tools_and_response
Passed	tests/unit/test_llm_client_and_providers.py::test_openai_provider_parsing_embedded_tool_call_and_wraps_errors
Passed	tests/unit/test_llm_client_and_providers.py::test_claude_provider_maps_messages_tools_and_usage
Passed	tests/unit/test_llm_client_and_providers.py::test_claude_provider_wraps_transport_errors
Passed	tests/unit/test_medical_tools.py::test_query_drug_formats_single_and_multi_term_results
Passed	tests/unit/test_medical_tools.py::test_query_drug_not_found_and_error
Passed	tests/unit/test_medical_tools.py::test_allergy_check_covers_match_and_no_match
Passed	tests/unit/test_medical_tools.py::test_dosage_and_advice_generation
Passed	tests/unit/test_medical_tools.py::test_submit_approval_success_empty_and_exception
Passed	tests/unit/test_medical_tools.py::test_fill_prescription_real_api_and_fallback
Passed	tests/unit/test_medical_tools.py::test_fill_prescription_sync_and_register_tools
Passed	tests/unit/test_models.py::test_drug_from_dict_coerces_fields_and_round_trips
Passed	tests/unit/test_models.py::test_drug_stock_rules_cover_expired_deleted_and_available
Passed	tests/unit/test_models.py::test_category_from_dict_handles_optional_and_invalid_integers
Passed	tests/unit/test_models.py::test_order_status_serialization_and_age
Passed	tests/unit/test_models.py::test_approval_status_optional_fields_and_age
Passed	tests/unit/test_planner_storage.py::test_storage_crud_round_trip[sqlite]
Passed	tests/unit/test_planner_storage.py::test_storage_crud_round_trip[file]
Passed	tests/unit/test_planner_storage.py::test_sqlite_storage_filters_and_bulk_delete
Passed	tests/unit/test_planner_storage.py::test_sqlite_storage_updates_existing_task
Passed	tests/unit/test_planner_storage.py::test_file_storage_filter_and_invalid_json
Passed	tests/unit/test_screening.py::test_symptom_service_standardizes_expands_and_scores
Passed	tests/unit/test_screening.py::test_screening_query_matches_filters_ranks_and_limits
Passed	tests/unit/test_screening.py::test_screening_query_rejects_empty_symptoms
Passed	tests/unit/test_screening.py::test_batch_screening_returns_one_result_per_query
Passed	tests/unit/test_screening.py::test_ranking_engine_applies_stock_allergy_age_and_sorting
Passed	tests/unit/test_screening.py::test_history_service_crud_filter_and_statistics
Passed	tests/unit/test_screening.py::test_config_service_validation_without_database_initialization
Passed	tests/unit/test_session_manager.py::test_session_manager_creates_caches_and_saves_agents
Passed	tests/unit/test_session_manager.py::test_session_manager_loads_existing_state
Passed	tests/unit/test_session_manager.py::test_session_manager_create_wrap_and_delete
Passed	tests/unit/test_symptom_db_service.py::test_synonym_lookup_and_expansion
Passed	tests/unit/test_symptom_db_service.py::test_drug_lookup_respects_stock_and_deleted_filters
Passed	tests/unit/test_upload_controller.py::test_upload_download_delete_lifecycle
Passed	tests/unit/test_upload_controller.py::test_upload_validation_errors
Passed	tests/unit/test_upload_controller.py::test_delete_reports_filesystem_error
Passed	tests/unit/test_upload_controller.py::test_allowed_extension_helper

Intelligent Medicine Pickup System - Coverage Report: 85%



Files

Functions

Classes

filter...

coverage.py v7.14.1, created at 2026-06-07 04:55 +0800

☐ hide covered

File ▲	statements	missing	excluded	coverage
agent__init__.py	2	0	0	100%
agent\engine__init__.py	26	5	0	81%
agent\engine\medical_agent.py	697	42	0	94%
agent\engine\workflows.py	126	13	0	90%
agent\llm__init__.py	3	0	0	100%
agent\llm\client.py	62	1	0	98%
agent\llm\providers\claude.py	43	9	0	79%
agent\llm\providers\openai.py	49	10	0	80%
agent\llm\schemas.py	50	5	0	90%
agent\memory__init__.py	3	0	0	100%
agent\memory\compressor.py	32	2	0	94%
agent\memory\manager.py	62	1	0	98%
agent\planner__init__.py	3	0	0	100%
agent\planner\models.py	183	4	0	98%
agent\planner\storage.py	258	56	0	78%
agent\services\symptom_service.py	24	3	0	88%
agent\session__init__.py	2	0	0	100%
agent\session\manager.py	68	1	0	99%
agent\subagents__init__.py	6	0	0	100%
agent\subagents\api.py	21	6	0	71%
agent\subagents\exceptions.py	2	0	0	100%
agent\subagents\extractor.py	189	16	0	92%
agent\subagents\form.py	763	23	0	97%
agent\subagents\models.py	31	2	0	94%
agent\subagents\symptom_extractor.py	2	0	0	100%
agent\tools__init__.py	4	0	0	100%
agent\tools\base.py	21	3	0	86%
agent\tools\executor.py	30	5	0	83%
Total	8746	1302	5	85%

<i>File ▲</i>	<i>statements</i>	<i>missing</i>	<i>excluded</i>	<i>coverage</i>
agent\tools\inventory.py	47	0	2	100%
agent\tools\medical.py	149	17	0	89%
agent\tools\registry.py	19	6	0	68%
agent\tools\report_generator.py	28	0	0	100%
api\approval_controller.py	243	109	0	55%
api\category_controller.py	94	22	0	77%
api\drug_controller.py	722	226	0	69%
api\health_controller.py	13	5	0	62%
api\order_controller.py	236	36	0	85%
api\ros_state_controller.py	41	5	0	88%
api\upload_controller.py	59	0	0	100%
auth__init__.py	4	0	0	100%
auth\audit.py	16	0	0	100%
auth\blueprint.py	368	159	0	57%
auth\constants.py	23	0	0	100%
auth\db.py	13	0	0	100%
auth\middleware.py	79	7	0	91%
auth\schema.py	114	15	0	87%
auth\tokens.py	24	0	0	100%
backend\init_db.py	40	7	3	82%
common\config.py	99	19	0	81%
common\exceptions.py	38	4	0	89%
common\utils__init__.py	5	0	0	100%
common\utils\cache.py	101	11	0	89%
common\utils\database.py	89	5	0	94%
common\utils\debug_logger.py	31	2	0	94%
common\utils\drug_helpers.py	57	20	0	65%
common\utils\drug_service.py	94	24	0	74%
common\utils\http_client.py	269	15	0	94%
common\utils\json_tools.py	22	2	0	91%
common\utils\logger.py	13	0	0	100%
common\utils\response.py	35	2	0	94%
common\utils\retry.py	22	1	0	95%
Total	8746	1302	5	85%

<i>File ▲</i>	<i>statements</i>	<i>missing</i>	<i>excluded</i>	<i>coverage</i>
common\utils\symptom_db_service.py	96	0	0	100%
common\utils\text_utils.py	71	0	0	100%
common\utils\validation.py	205	6	0	97%
database\approval_manager.py	147	5	0	97%
database\connection.py	84	0	0	100%
database\helpers.py	57	6	0	89%
database\models\approval.py	52	2	0	96%
database\models\category.py	37	4	0	89%
database\models\drug.py	93	10	0	89%
database\models\order.py	31	2	0	94%
database\pharmacy_client.py	178	21	0	88%
ros_integration__init__.py	28	13	0	54%
ros_integration\bridge.py	131	51	0	61%
ros_integration\config.py	38	0	0	100%
ros_integration\error_handler.py	143	7	0	95%
ros_integration\health_monitor.py	76	6	0	92%
ros_integration\message_adapter.py	75	4	0	95%
ros_integration\node_manager.py	114	21	0	82%
ros_integration\state_store.py	63	0	0	100%
ros_integration\state_subscriber.py	155	58	0	63%
ros_integration\task_publisher.py	233	92	0	61%
screening__init__.py	1	0	0	100%
screening\models__init__.py	4	0	0	100%
screening\models\screening_config_model.py	35	1	0	97%
screening\models\screening_history_model.py	31	1	0	97%
screening\models\symptom_model.py	42	3	0	93%
screening\routes__init__.py	2	0	0	100%
screening\routes\screening_routes.py	150	31	0	79%
screening\services__init__.py	5	0	0	100%
screening\services\config_service.py	135	5	0	96%
screening\services\history_service.py	62	6	0	90%
screening\services\ranking_engine.py	67	5	0	93%
screening\services\screening_service.py	79	11	0	86%
Total	8746	1302	5	85%

<i>File ▲</i>	<i>statements</i>	<i>missing</i>	<i>excluded</i>	<i>coverage</i>
screening\services\symptom_service.py	57	5	0	91%
Total	8746	1302	5	85%

6 empty files skipped.

coverage.py v7.14.1, created at 2026-06-07 04:55 +0800